A-A-54368
15 April 1991

COMMERCIAL ITEM DESCRIPTION

DEFIBRILLATOR AND CARDIOSCOPE

(Three Lead, Portable, with Charger)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a cardiac care system suitable for use on patients experiencing cardiac crisis and for examining the electrical activity of the heart, with a means to produce a permanent record of the heart activity in the three standard ECG lead formats.

Salient characteristics:

The unit shall be an integral or modular portable ECG monitor/recorder and defibrillator which functions from self-contained batteries. Units which are modular in design and can be separated shall consist of only two separate components: a defibrillator and a monitor/recorder. Each component shall be independently powered and shall securely interconnect into a physically-contiguous unit without the use of tools, without the need for auxiliary cables, and in such a manner as not to cause excessive stress on the electrical connectors.

The defibrillator and cardioscope shall include batteries, separate external adult and external pediatric paddles, synchronized cardioversion capability, and accessories as specified herein. Provision shall be made for out-of-the-way paddle storage. The unit shall also include a battery charger which shall be separate from the monitor/recorder and defibrillator.

AMSC/NA FSC 6515

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The unit must be designed in such a way as to respond to a variety of cardiac crises and/or trauma situations that may be encountered by emergency medical teams operating in conjunction with an ambulance or rescue vehicle. Typical environments include:

1. A wet swimming pool side.
2. A roadside during a rainstorm.
3. A smoky environment near a fire.
4. A dusty environment at a construction site.
5. Air/Sea rescue situation (carried on the helicopter, in which the unit is subjected to sea water drawn up by "prop wash").

The system as either independent modules and/or as an integral unit shall comply with applicable sections and requirements of:

* American National Standards Institute/Association for the Advancement of Medical Instrumentation DF2, ANSI/AAMI DF2 - American National Standard for Cardiac Defibrillator Devices, with the exception of paragraph 3.2.1.2, Energy Range.

* Underwriters Laboratories Standard 544, UL 544 - Standard for Safety of Electric Medical and Dental Equipment.

* National Fire Protection Association 99, NFPA 99 - Standard for Health Care Facilities.

* American Heart Association, AHA publication titled, "Recommendations for Standardization of Leads and of Specifications for Instruments in Electrocardiography and Vectorcardiography".

* Association for the Advancement of Medical Instrumentation, AAMI ECG 5/83 Standard for ECG Connectors.

The defibrillator system shall be capable of full operation while still providing environmental protection and shall comply with environmental conditions as follows:

Atmospheric Pressure:	500 mm Hg - 700 mm Hg
Relative Humidity (R.H.):	95 percent (non-condensing)
Temperature:	
Operational.....	32^0F to 114^0F
Storage.....	-23^0F to 149^0F

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All units shall be easily cleaned with routine hospital cleansing solutions, adequately protected from moisture, and be designed to withstand normal abuse, such as shock and impact, frequently encountered in circumstances outside the normal hospital environment. The unit must be capable of being fully operated while still providing environmental protection from moisture, dust or mud.

The unit shall be capable of withstanding the drip proof test specified herein without damage and shall operate properly in all modes following the test (see Environmental Safety/Drip-Proof Test). All controls must be immediately accessible (no cover) during all steps of the test.

The total weight of the ECG monitor/recorder and defibrillator including the batteries, paddles, etc., shall not exceed 30 pounds.

ECG monitor/recorder. The ECG monitor/recorder shall have the following characteristics:

a. The instrument shall be capable of providing an ECG through the defibrillator paddles or via a three (3) lead patient cable having a six (6) pin cable connector, configured in accordance with with AAMI ECG-5/83 Standard for ECG connectors.

b. Cardioscope, recorder, and ECG amplifier performance requirements shall be in accordance with Standard for Cardiac Monitors, Heart Rate Meters and Alarms, AAMI, unless otherwise specified.

c. The patient cable leads supplied with the monitor shall be color coded in conformance with AHA publication, "Recommendations for Standardization of Leads and of specifications for Instruments in Electrocardiography and Vectorcardiography".

d. The patient cable shall shielded and have an approximate minimum length of 10 feet. Individual lead length shall be approximately 3 feet. Total cable length with patient leads should not exceed 13 feet.

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Cardioscope-display. The cardioscope display shall have the following characteristics:

a. Shall be non-fade type display measuring no less than 36 mm high by 60 mm wide.

b. Unit shall have either a freeze control or a delayed recording capability for detailed analysis of ECG arrhythmias.

c. Shall display a signal with a measurable low frequency cutoff point (-3 db down) at 0.5 Hz, and a high frequency cutoff point (-3 db down) anywhere between 35 to 50 Hz inclusive.

d. Shall have sweep speed of 25 mm/second.

e. Shall have an easily adjustable audible systole indicator.

f. Shall have a lead selector switch enabling the operator to select lead I, II, and III without switching patient electrode sites.

g. ECG gain shall be continuously adjustable or discrete step mode.

ECG amplifier. The ECG amplifier shall have the following characteristics:

a. Shall be protected from defibrillator counter shock up to a level of 400 joules.

b. Shall have a minimum common mode rejection ratio (CMRR) of 80 db minimum with respect to chassis ground with 5K ohms imbalance at 60 Hz.

c. Shall have an electrically isolated patient input circuit in accordance with the latest edition of the American National Standard Safe Current Limits for Electromedical Apparatus, ANSI/AAMI SCL.

d. Shall have a 1 mV reference signal for ECG calibration.

e. A low paper indicator shall be provided, either by the instrument or by a mark on the paper no less than three feet from the end of the roll.

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Recorder. The recorder shall have the following characteristics:

a. Frequency response shall conform at a minimum to the specifications given for the cardioscope display.

b. Shall have a paper speed of 25 mm/second. Paper speed shall be accurate to within plus or minus 5 percent.

c. ECG paper shall have a minimum grid width of 40mm and contain at least 50 feet per roll.

d. A low paper indication shall be provided, either by the instrument or in the form of a mark on the paper.

e. The recorder must be an integral part of the monitor section of the defibrillator/monitor instrument and not a stand-alone instrument requiring electrical cabling inter-connection to provide patient data.

Defibrillator. The defibrillator shall have the following characteristics:

a. The defibrillator performance requirements shall be in conformance with American National Standard for Cardiac Defibrillator Devices, ANSI/AAMI DF2, unless otherwise specified.

b. The defibrillator pulse shape shall be a clinically proven one such as a Lown, Edmark or Trapezoidal, as defined in AAMI DF2.

c. The maximum selectable delivered energy of the defibrillator shall lie in the range 320 to 400 joules (J) inclusive. The defibrillator shall provide a selection of at least six energy outputs within the range of 5 J to maximum energy when delivered into a 50 ohm resistive load. At least three (3) of these outputs shall be less than 110 joules.

d. The delivered pulse energy shall be within 4 J or plus or minus 15 percent of the selected energy on the energy level indicator, whichever is greater, when discharged into a 50 ohm resistive load.

e. The defibrillator shall be capable of delivering 10 pulses with maximum energies into a 50 ohm resistive load within 5 minutes, at 32^0F.

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f. Paddles shall conform to American Heart Association dimensions and shall be capable of ECG pick-up through the paddle surface. Both paddles shall be provided with discharge buttons and both must be activated simultaneously to discharge the unit. One paddle shall be provided with a remote charge button. The external pediatric paddles shall provide at least a 15 square centimeter unlimited contact area per electrode.

g. Shall have a maximum charge time (to maximum energy setting) of 12 seconds or less at 82.5°F.

h. The unit shall have a synchronizer with visual indication on the cardioscope to identify the synchronizer trigger point when the cardioversion mode of the defibrillator is activated. Unit must switch back to difibrillation mode after each cardioversion attempt.

i. Shall provide a visual means of determining selected energy by a calibrated knob setting, digital display, or analog meter.

j. Shall provide a visual ready indicator and audible tone to indicate when defibrillator is fully charged and ready for discharge.

Electrical Characteristics. The ECG monitor/recorder and defibrillator shall operate from self-contained batteries. The system shall be capable of providing all functions specified herein. Defibrillator and ECG monitor/recorder systems that can be used independently shall be independently powered, each containing internal battery power source.

a. The defibrillator/monitor-recorder charging system shall operate over the full tolerance range of one or more of the following line voltage and frequency combinations as specified by the customer requisition:

* 115 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
* 115 VAC plus or minus 10 percent, 60 plus or minus 2 Hz
* 230 VAC plus or minus 10 percent, 50 plus or minus 2 Hz

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c. ECG paper shall have a minimum grid width of 40mm and contain at least 50 feet per roll.

d. A low paper indication shall be provided, either by the instrument or in the form of a mark on the paper.

e. The recorder must be an integral part of the monitor section of the defibrillator/monitor instrument and not a stand-alone instrument requiring electrical cabling inter-connection to provide patient data.

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a. The defibrillator performance requirements shall be in conformance with American National Standard for Cardiac Defibrillator Devices, ANSI/AAMI DF2, unless otherwise specified.

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c. The maximum selectable delivered energy of the defibrillator shall lie in the range 320 to 400 joules (J) inclusive. The defibrillator shall provide a selection of at least six energy outputs within the range of 5 J to maximum energy when delivered into a 50 ohm resistive load. At least three (3) of these outputs shall be less than 110 joules.

d. The delivered pulse energy shall be within 4 J or plus or minus 15 percent of the selected energy on the energy level indicator, whichever is greater, when discharged into a 50 ohm resistive load.

e. The defibrillator shall be capable of delivering 10 pulses with maximum energies into a 50 ohm resistive load within 5 minutes, at 32^0F.

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h. The unit shall have a synchronizer with visual indication on the cardioscope to identify the synchronizer trigger point when the cardioversion mode of the defibrillator is activated. Unit must switch back to difibrillation mode after each cardioversion attempt.

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j. Shall provide a visual ready indicator and audible tone to indicate when defibrillator is fully charged and ready for discharge.

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a. The defibrillator/monitor-recorder charging system shall operate over the full tolerance range of one or more of the following line voltage and frequency combinations as specified by the customer requisition:

* 115 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
* 115 VAC plus or minus 10 percent, 60 plus or minus 2 Hz
* 230 VAC plus or minus 10 percent, 50 plus or minus 2 Hz

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Dual voltage and hertz on all units are desirable but not mandatory.

b. Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

c. The defibrillator/monitor-recorder shall be capable of full operation while powered by internal batteries for the following functions:

1. Cardiac monitor - minimum of 2.25 consecutive hours during continuous monitor operation.

2. Cardiac monitor-recorder - minimum of 45 consecutive minutes during use of the monitor and strip chart recorder or any linear combination thereof.

3. Defibrillator:

(a) Thirty defibrillator counter shocks at maximum energy into a 50 ohm load at an ambient temperature of 82.5°F.

(b) Charge time to maximum energy output shall not exceed 12 seconds at 82.5°F.

(c) Ten defibrillator counter shocks at maximum energy into a 50 ohm load at an ambient temperature of 32°F within a period of five minutes.

d. Battery Charging System:

1. Charge time for fully depleted batteries to fully charged state shall not exceed sixteen hours. The unit shall have a level indicator to show the level to which the batteries have been recharged after the sixteen hours.

2. A charging lamp/indicator shall be provided for each charge circuit which is illuminated during the periods when the battery charger is functioning.

3. A low voltage indicator shall be provided on the front of the defibrillator and monitor-recorder for each battery used.

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4. The defibrillator and monitor-recorder shall allow rapid interchange (operator installable/cassette type) of battery packs, and be provided with two spare battery packs in addition to the batteries supplied with the unit.

5. Battery charger shall be supplied with a UL approved three-wire ground line cord with hospital grade plug.

Accessories. Shall be provided but not limited to the following accessories:

a. Two (2) each patient cables.

b. One (1) spare stylus, if applicable (with replacement tool, if required).

c. One (1) ECG ruler.

d. Three (3) pre-gelled electrodes.

e. Two (2) spare batteries of each type used.

f. Three (3) packages recording paper, marked with Federal/National Stock Number.

g. One (1) tube electrode jelly.

h. One (1) accessory bag.

The unit shall be of a rugged construction capable of withstanding hazards of normal use in a hospital setting as well as those found in field and exercise conditions.

Operation and Service Data Manuals shall be furnished as follows:

a. Information Content:

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language and all schematics

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and wiring diagrams will use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc, pertaining to the equipment and issued subsequent to the printing of he original instructions shall also be included.

These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

b. Quantities, Distribution and Schedules:

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of 2 service manuals need to be supplied to each consignee. However, 2 operation manuals still need to be supplied with each unit.

Workmanship. The defibrillator system shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One battery operated defibrillator/monitor-recorder, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one defibrillator system.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105 with an AQL of 1.5 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one defibrillator system.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

Strain Relief test (Battery Charger Line Cord).

Gradually apply a 35 lb. pull in a direction that would be the most likely to pull the cord from the enclosure and maintain for one minute. The strain relief is not acceptable if there is such movement of the cord as to indicate that stress is being transmitted to terminals, splices or interior wiring.

Air Clearance Spacing test.

Measurements shall be made, using an appropriate measuring instrument, to verify isolated patient connected circuitry is spaced at a minimum of 1/8 inch from non-isolated circuitry.

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Leakage current test:

Defibrillator

1. Chassis Leakage:
Leakage current measured between any exposed conductive surface and ground shall not exceed 100 microamperes.

2. Paddle Leakage:
Leakage current measured between an external paddle electrode and ground shall not exceed 100 microamperes.
Leakage current measured between an internal paddle electrode and ground shall not exceed 50 microamperes.

3. Conditions:
Leakage current measurements will be made under all of the following conditions.

a) Power on and off.
b) Defibrillator in both charged and uncharged state.

ECG Monitor/Recorder

1. Chassis Leakage:
Leakage current measured between any exposed conductive surface and ground shall not exceed 100 microamperes

2. Lead leakage (source current)leakage current measured between each lead and ground, all leads ground, and between any two leads shall not exceed 10 microamperes.

3. Lead leakage (sink current):
Sink current measured between each load at a potential of 120V and ground shall not exceed 20 microamperes. This test shall be performed under the conditions listed above except the ground remains intact.

Test Device

All leakage current measurement will be performed with a test device equipped with an AAMI Standard test load.

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Defibrillator Paddle Insulation test.

a. Apply aluminum foil to the entire gripping surface of one paddle with sufficient pressure to make it conform to the surface.

NOTE: The gripping surface for verification purposes is defined as the approximately cylindrical area around each handle where the operator's fingers grasp the paddle, but not beyond any barrier or curvature that is obviously intended to limit hand placement, including areas one inch in diameter centered on any control located on the paddle, representing the location(s) of the operator's thumb and/or finger.

b. With the paddles connected to the defibrillator, connect the high potential lead of a 60 Hz (or 50 Hz, if preferred) high potential test source to the aluminum foil. Connect low-potential or ground lead to the paddle electrode. Slowly adjust the peak voltage amplitude of the test source to equal 1.5 times the maximum voltage occurring on the defibrillator energy storage capacitor. For a sine wave this corresponds to an rms value of 1.06 times the capacitor voltage. Maintain this voltage for 1 minute. Repeat this test with the low-potential lead connected to the other paddle electrode.

c. There shall be no voltage breakdown, and the 60 Hz leakage current shall not exceed 250 microamperes peak (177 microamperes rms); if a 50 Hz test source is used, the current shall not exceed 210 microamperes peak (148 microamperes rms).

NOTE: Any leakage current that flows when the high-potential lead is disconnected from foil, but the output voltage is set as described in paragraph b above, represents internal leakage of the test apparatus and may be subtracted from the total current reading. Alternatively, an external ammeter may be placed in the ground circuit to eliminate this internal leakage and to enhance resolution, if necessary.

Repeat steps as described in paragraphs a, b, and c above, using the other paddle grip.

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ECG Monitor-Recorder Sweep/Chart Speed test.

Measure the monitor sweep speed and the recorder paper speed by applying a signal of known frequency to the patient cable. Monitor sweep speed shall be within plus or minus 10 percent of 25 mm/sec. and recorder chart speed within plus or minus 5 percent of 25 mm/sec.

ECG Amplifier Calibration test.

The accuracy of the 1 mV internal calibration pulse shall be measured by comparing the amplitude of this pulse to that of a calibrated 1 mV pulse applied to the patient cable from a pulse generator. It shall be accurate to plus or minus 15 percent.

ECG Amplifier Common Mode Rejection test.

Common mode rejection is measured by comparing the amplitude of a 1 mV differential signal applied to the monitor to the amplitude of a higher voltage sinusoidal signal of 60 Hz applied between the junction of the leads and ground. Measured common mode rejection shall be at least 80 dB with respect to chassis ground with 5K ohms imbalance between leads.

Defibrillator Overload Protection test.

Overload protection shall comply with those requirements specified in ANSI/AAMI EC 13.

ECG Monitor and Recorder Frequency Response test.

Apply a 1 mV peak-to-peak differential sinusoidal signal to the patient cable, and vary the frequency. The bandwidth is defined by the upper and lower frequencies at which the trace deflection amplitude decreases to 0.707 times the value. This bandwidth maximum shall be 0.5 Hz to anywhere between 35 to 50 Hz inclusive, on the monitor and recorder, real time mode.

Defibrillator Energy Accuracy test.

Connect the defibrillator paddles to a defibrillator analyzer equipped with a 50 ohm resistive load. Charge the defibrillator to a preselected energy level, then discharge into the analyzer. Repeat for each available energy level setting.

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Defibrillator Charge Time test.

Using a stopwatch, measure the time between activation of the charge control and full charge indication for the first and last of 10 sequential charges and discharges all at maximum energy. This time shall not exceed 12 seconds.

Defibrillator Discharge Pulse Rate test.

Apply the paddle electrodes to a defibrillator tester and select the maximum energy level. Simultaneously initiate the charge and a timer.

Immediately upon completion of the charge, discharge the defibrillator.

The defibrillator shall be capable of delivering 10 pulses at maximum energy, into a 50 ohm resistive load within 5 minutes, at 32°F.

Defibrillator Overload Protection test.

Overload protection shall comply with those requirements specified in ANSI/AAMI EC 13, except the defibrillator energy level shall be 400J.

Synchronized Discharge test.

Connect a signal generator to the ECG monitor. Measure the time between the leading edge of square pulse and the leading edge of the defibrillator's synchronized discharge pulse. This time can be displayed automatically on a defibrillator analyzer if so equipped, or measured directly from an oscilloscope, and shall be within 30 milli-seconds.

Battery Capacity test.

Battery Capacity will be tested by verifying 2.25 hours of continuous cardioscope monitoring or 45 minutes of continuous recording of a heart rate of 60 BPM. Defibrillator battery capacity will be tested by verifying 30 discharges at maximum energy into a 50 ohm load at an ambient temperature of 82.5°F.

Environmental Safety/Drip-Proof test.

The contractor shall be responsible for the performance of drip-proof test as follows:

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Step 1. Place item in its normal operating position, with patient cable and defibrillator paddles attached, on a platform or tilt table in a manner such that it can be oriented horizontally and in any of four directions forward, backward, and to each side to the extent of 15 degrees from the horizontal plane.

Step 2. Attach a spray nozzle with a uniformly distributed full cone spray pattern to a flexible hose and using a container calibrated in liters or an equivalent method, determine the rate of flow out of the nozzle. The requirements for water flow shall be 4.3 gallons per hour for each square foot of area covered by the spray 3 feet below the nozzle.

Step 3. Position the nozzle 3 feet above the item so that the spray is directed downward onto it. The item shall be subjected to a uniform spray over the entire surface being tested. During the test, the item shall not be operated, but all controls shall be immediately accessible to the operator: i.e., uncovered, with the exception of the recorder, which may be protected.

Step 4. Direct the spray onto the top of the item (with platform or tilt table horizontal) for at least eight minutes. Turn off spray four minutes.

Step 5. Direct the spray downward onto the item for at least eight minutes with the platform or table tilted such that the item is oriented 15 degrees from the horizontal in the forward position. Turn off spray four minutes.

Step 6. Direct the spray downward onto the item for at least eight minutes with the platform or table tilted such that the item is oriented 15 degrees from the horizontal in the backward position. Turn off spray four minutes.

Step 7. Direct the spray downward onto each remaining side of the item for at least eight minutes with the platform or table tilted such that the side being sprayed is oriented 15 degrees from the vertical. Turn off spray for four minutes.

Step 8. Return the item to the horizontal and perform a satisfactory operation test.

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Humidity test.

Manufacturer shall supply the following, verifying compliance with the test described below:

a. Copy of test procedure as specified below in test method.

b. Unit type.

c. Serial number.

d. Manufacturer and model number of chamber used.

e. Leakage current test results after 48 hours.

f. Energy accuracy test results after 48 hours.

g. Test Method

1. Place the unit in test chamber such that leakage current and delivered energy accuracy tests can be performed external to the chamber.

2. Adjust the temperature of the chamber to 82°F.

3. Permit sufficient time for the temperature "inside the unit" to stabilize prior to proceeding.

4. After temperature stabilization, adjust the R.H. of the chamber to 95 (plus or minus 5) percent and permit sufficient time for humidity stabilization.

5. After 48 hours of conditioning, source and sink leakage current tests and energy accuracy tests are to be performed. The unit is in compliance if leakage currents and energy accuracy are within the specified limits.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6515-01-304-7944

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6515-5005
Location: DEF3CID

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-54368	2. DOCUMENT DATE (YYMMDD) 15 April 1991

3. DOCUMENT TITLE
DEFIBRILLATOR AND CARDIOSCOPE (Three-Lead, Portable, with Charger)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSTE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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